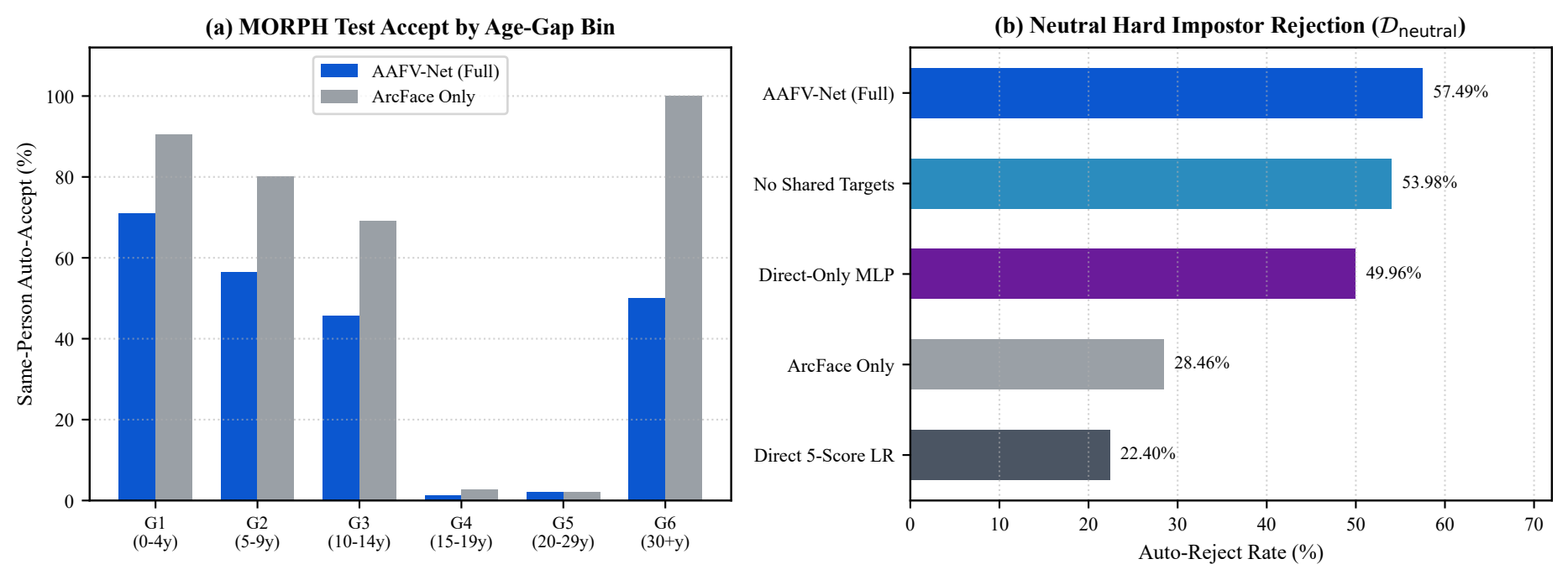

(a) MORPH Test Accept by Age-Gap Bin
AAFV-Net (Full)
ArcFace Only
Same-Person Auto-Accept (%)
0
20
40
60
80
100
G1
(0-4y)
G2
(5-9y)
G3
(10-14y)
G4
(15-19y)
G5
(20-29y)
G6
(30+y)
(b) Neutral Hard Impostor Rejection (D_{neutral})
AAFV-Net (Full)
57.49%
No Shared Targets
53.98%
Direct-Only MLP
49.96%
ArcFace Only
28.46%
Direct 5-Score LR
22.40%
0
10
20
30
40
50
60
70
Auto-Reject Rate (%)